SAM 3D: Breaking the 3D Data Barrier for Object Reconstruction in the Wild

Anonymous CVPR submission

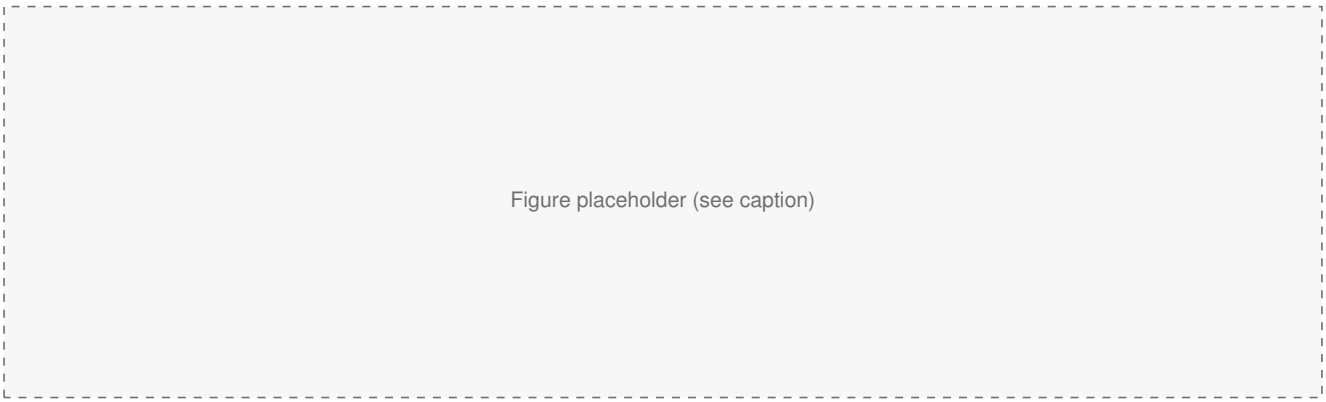


Figure 1. **3D reconstruction of objects in natural images.** SAM 3D predicts the shape, texture and layout of each prompted object from a single image, including the parts hidden by occlusion or cut off by the frame, and the objects of one image can be composed into a scene. **[TODO: Figure plan: 3 to 4 natural images with clutter and occlusion (left: image with object prompts), and for each the reconstructed objects rendered from a new viewpoint and composed into the scene with their predicted layout (right); one zoom-in on an occluded object showing its completed hidden part.]**

Abstract

We present SAM 3D, a generative model for visually grounded 3D object reconstruction that predicts the geometry, texture and layout of an object from a single image. SAM 3D is built for natural images, where objects are occluded and surrounded by clutter, and where the context of a scene often says more about an object than its visible pixels. Reconstruction in this setting is limited less by models than by data: 3D ground truth for objects in natural images is scarce, while the synthetic assets that are plentiful show no real scenes. We break this data barrier with a human- and model-in-the-loop pipeline that annotates object shape, texture and pose, and that yields visually grounded 3D reconstruction data at unprecedented scale [TODO: number of objects and images]. We learn from this data with a multi-stage training framework that combines synthetic pretraining with real-world alignment. SAM 3D outperforms recent work and wins human preference tests on real-world objects and scenes by at least 5:1. We will release our code, model weights, an online demo and a new challenging benchmark for in-the-wild 3D object reconstruction.

1. Introduction

Take a photograph of a living room and ask for the 3D shape of the chair in it. Its legs are hidden behind a coffee table, its back is cut off by the edge of the frame, and a cushion covers half of its seat. A person still sees a complete chair with four legs, because the room, the table and the visible parts tell them what the hidden parts must be. Single-image 3D reconstruction has become remarkably good at objects shown alone on a clean background [TODO: cite recent single-image 3D reconstruction and generation], yet the chair in the living room is the common case in natural images (Fig. 1).

Recent models owe their quality to large collections of synthetic 3D assets, which can be rendered from any viewpoint to supervise reconstruction [TODO: cite]. These renderings show objects in isolation, without the occlusion, clutter and context of real scenes. The main reason models do not close this gap is data. 3D ground truth for objects in natural images is expensive to obtain, because it requires a 3D artist or a scanner for every object [TODO: cost per object, if known], and existing real-world 3D datasets are small or limited to a few categories [TODO: cite]. Synthetic data is plentiful but not natural, while natural images

are plentiful but have no 3D. We call this the 3D data barrier.

Annotating 3D at scale needs a different division of labour. People are slow at modelling a 3D shape, but they can quickly judge whether a 3D object matches what they see in an image. Models are the opposite: they produce 3D candidates in seconds but cannot tell when a candidate is wrong. We combine the two in a human- and model-in-the-loop pipeline in which models propose shapes, textures and poses for objects in natural images and annotators [TODO: what annotators do: select, rank, correct or reject candidates] (Sec. 2.3). Each round of annotation trains a better model, whose candidates make the next round faster and more accurate, so the pipeline yields visually grounded 3D reconstruction data at unprecedented scale [TODO: number of objects, images and categories].

We train SAM 3D, a generative model that predicts the geometry, texture and layout of an object from a single image and a prompt [TODO: prompt type, e.g., mask, box or point], on this data with a multi-stage framework. Synthetic pretraining gives the model a broad prior over 3D shape and appearance, and real-world alignment on the annotated data teaches it to apply this prior under occlusion and clutter [TODO: stages and objectives of real-world alignment]. Because the model is generative, it completes hidden parts with plausible geometry that agrees with the visible ones.

Unlike recent methods that are trained only on synthetic assets [TODO: cite closest work], SAM 3D learns from objects in natural images, so occlusion, clutter and context are part of its training distribution. The closest work is [TODO: name it], which differs from ours in [TODO: one checkable difference: training data, outputs (layout) or input domain].

SAM 3D outperforms recent work by a wide margin: in human preference tests on real-world objects and scenes, annotators prefer its reconstructions at least five times as often as those of every baseline [TODO: name the baselines; confirm that 5:1 holds for each]. To measure progress in this setting, we introduce a new benchmark for in-the-wild 3D object reconstruction [TODO: benchmark name and size], and we will release our code, model weights and an online demo. Our results show that the main obstacle to 3D reconstruction in natural images was data, and that people and models annotating together can remove it.

2. SAM 3D: Learning 3D Reconstruction from Natural Images

Given a natural image and a prompt that selects one object, SAM 3D predicts the object's geometry, texture and layout in the camera frame (Fig. 2a). The method has three parts. We first define the task and the generative model (Secs. 2.1 and 2.2). We then describe the human- and model-in-the-loop pipeline that produces real-world training data (Sec. 2.3), and the multi-stage framework that trains the model on synthetic and real data (Sec. 2.4).

2.1. Task: visually grounded 3D reconstruction

The input is an image I and a prompt p that identifies one object in it [TODO: prompt type and format]. The output is the object's shape S, its texture T and its layout $\ell = (\mathbf{R}, \mathbf{t}, s)$, the rotation, translation and scale that place the object in the camera frame of I [TODO: representations of S and T, and the layout parameterization]. The reconstruction is visually grounded: the visible parts of S and T must match the pixels of I, and the hidden parts must be consistent with the object and its context. Predicting ℓ together with S and T lets the reconstructions of several objects in one image be composed into a scene.

2.2. A generative model of objects in context

Occluded and truncated parts admit many plausible completions, and a regression model trained on them predicts their average, which is blurred and often not a valid object. SAM 3D is therefore generative: it learns the conditional distribution

$$p_\theta(S, T, \ell \mid I, p), \tag{1}$$

from which we sample one reconstruction per object at inference. [TODO: Architecture: how I and p are encoded, whether geometry, texture and layout are generated jointly or in stages, the generative formulation (e.g., diffusion or flow matching) and its latent space.] Conditioning on the full image, and not only on a crop of the object, lets the model use the scene around the object as a recognition cue [TODO: confirm that the full image is used as context].

2.3. Human- and model-in-the-loop data engine

The model in Eq. (1) needs pairs of natural images and 3D objects, which no existing dataset provides at scale. Creating each 3D object by hand is too slow, so we let models do the creation and let people do the judging. In round k of the pipeline, the current model f_k proposes candidate reconstructions for objects in natural images [TODO: image sources and how objects are selected], and annotators [TODO: the annotation operation: select the best candidate, rank candidates, edit pose, or reject] (Fig. 2b). Shape, texture and pose are annotated [TODO: jointly or in separate steps; tools used for pose alignment]. The accepted annotations form the dataset $\mathcal{D}_k$, on which we train the next model,

$$f_{k+1} = \mathrm{Train}\big(f_k, \textstyle\bigcup_{j \le k} \mathcal{D}_j\big). \tag{2}$$

The loop improves both sides at once: a better model proposes better candidates, so more objects are accepted per unit of annotation time, and the accepted objects are harder ones that the previous model could not reconstruct. [TODO: How objects with no acceptable candidate are handled; quality control; number of rounds; final dataset statistics.]

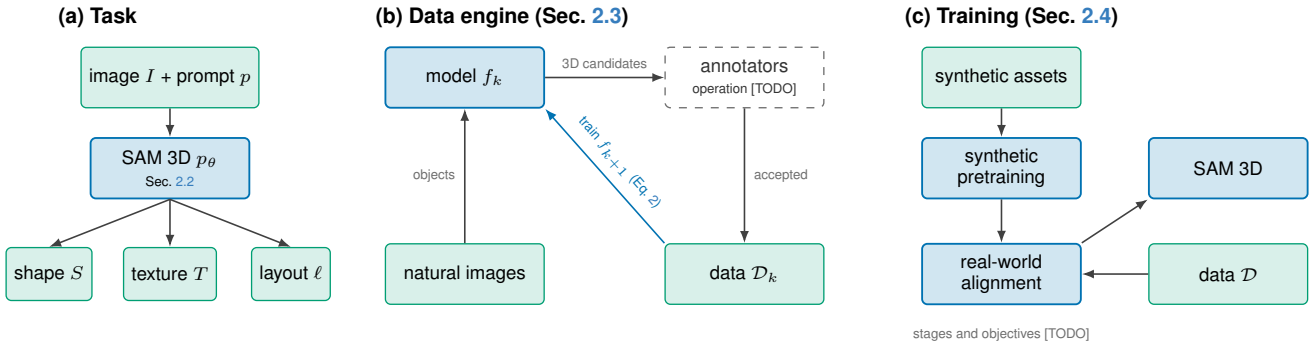


Figure 2. **Overview.** (a) Given an image and a prompt for one object, SAM 3D samples its shape, texture and layout. (b) In each round of the data engine, the current model proposes 3D candidates for objects in natural images, annotators judge them, and the accepted annotations train the next model. (c) Synthetic pretraining gives a prior over 3D shape and texture; real-world alignment on the data-engine data adapts it to natural images. Blue parts are ours, green is data; dashed parts are not yet specified.

2.4. Multi-stage training

Synthetic 3D assets cover many shapes and materials but show no real scenes, while $\mathcal{D}$ shows real scenes but covers fewer objects. We use each for what it provides (Fig. 2c). In synthetic pretraining, SAM 3D learns a prior over 3D shape and texture from rendered synthetic assets [TODO: assets, rendering and the pretraining objective]. In real-world alignment, it adapts this prior to natural images by training on $\mathcal{D}$ [TODO: the stages of alignment and their objectives; if annotator choices are used as preferences, state the objective]. [TODO: Loss for each stage, e.g., $\mathcal{L} = \ldots$, and what is frozen or updated in each stage.]

Relation to synthetic-only training. The architecture and the pretraining stage resemble those of methods trained only on synthetic assets; the difference is the data that follows. Real-world alignment exposes the model to the occlusion, clutter and context that synthetic renderings lack, and it is the data engine that makes this stage possible at scale.

3. Experiments

Our experiments answer four questions. Do people prefer the reconstructions of SAM 3D over those of recent work on real-world objects and scenes (Sec. 3.2)? Does SAM 3D also improve geometry, texture and layout metrics (Sec. 3.3)? How much does each training stage contribute (Sec. 3.4)? How does quality change with the amount of data from the data engine (Sec. 3.5)?

3.1. Setup

Benchmark. Existing benchmarks show objects mostly in isolation [TODO: cite], so we introduce a new benchmark for in-the-wild 3D object reconstruction [TODO: name, number of images and objects, how ground truth was obtained, and how it is kept separate from the training data].

Table 1. **People prefer SAM 3D at least 5:1.** Win rate of SAM 3D against each baseline in pairwise human preference tests on real-world objects and scenes.

SAM 3D vs.	Objects	Scenes
Baseline 1	?	?
Baseline 2	?	?
Baseline 3	?	?

[TODO: Other benchmarks used, if any.]

Metrics. Our main metric is human preference: annotators see the image and two reconstructions and choose the better one [TODO: protocol: number of raters, views shown, ties, objects versus scenes]. We also report [TODO: geometry metrics, e.g., Chamfer distance ↓ and F-score ↑; texture metrics; layout metrics, e.g., rotation error and 3D IoU].

Baselines. We compare with [TODO: recent single-image 3D reconstruction and generation methods; for scenes, methods that predict layout], using official code and checkpoints [TODO: confirm]. Baselines that do not predict layout are given [TODO: how layout is provided or estimated for them].

Implementation. [TODO: Model size, training data per stage, optimizer, steps, hardware, sampling settings.]

3.2. Human preference

Annotators prefer SAM 3D over every baseline by at least 5:1 on both real-world objects and scenes (Tab. 1). [TODO: Name the baseline with the smallest margin and the largest one, with their win rates.] The margin is largest on [TODO: the subset, e.g., heavily occluded or small objects], where

Table 2. **SAM 3D improves geometry, texture and layout.** Results on our in-the-wild benchmark [TODO: and others]. Best in **bold**.

Method	CD ↓	F1 ↑	Texture ?	Layout ?
Baseline 1 ?	?	?	?	?
Baseline 2 ?	?	?	?	?
Baseline 3 ?	?	?	?	?
SAM 3D	?	?	?	?

Table 3. **Both stages are needed.** Ablations on our benchmark; each row changes one part of the training of SAM 3D.

Training	Win rate ↑	CD ↓	Layout ?
Full (pretraining + alignment)	?	?	?
Synthetic pretraining only	?	?	?
Real data only, no pretraining	?	?	?
Alignment without stage ?	?	?	?

baselines trained on isolated objects [TODO: describe their typical failure], while SAM 3D uses the surrounding scene to complete the object (Fig. 3).

3.3. Quantitative evaluation

[TODO: State the finding first, e.g., SAM 3D outperforms all baselines on geometry and layout] (Tab. 2), with [TODO: headline margin]. [TODO: Explain the gain with reference to the method, e.g., which errors of the baselines come from training only on isolated objects.] [TODO: If automatic metrics and human preference disagree for some baseline, report it here and explain why.]

3.4. Ablation study

Each ablation in Tab. 3 answers one question about the training framework (Sec. 2.4). *Does real-world alignment matter?* Training on synthetic data only [TODO: effect], mostly on occluded objects. *Does synthetic pretraining matter?* Training on $\mathcal{D}$ alone [TODO: effect], mostly on [TODO: e.g., rare shapes]. *Is every alignment stage needed?* Removing [TODO: stage] [TODO: effect].

3.5. Scaling with the data engine

Quality improves steadily with the amount of data from the data engine (Fig. 4) [TODO: confirm the trend and state it with the slope or the gain from first to last round]. [TODO: Annotation throughput per round: objects accepted per annotator hour, showing that better models make annotation faster] (Sec. 2.3).

3.6. Limitations

[TODO: Concrete failure modes with a cause and a figure. Candidates to verify are listed in the report: objects absent from both synthetic and annotated data, thin or transparent structures, layout errors under strong perspective, and dependence of the data engine on annotator judgement for hidden parts.]

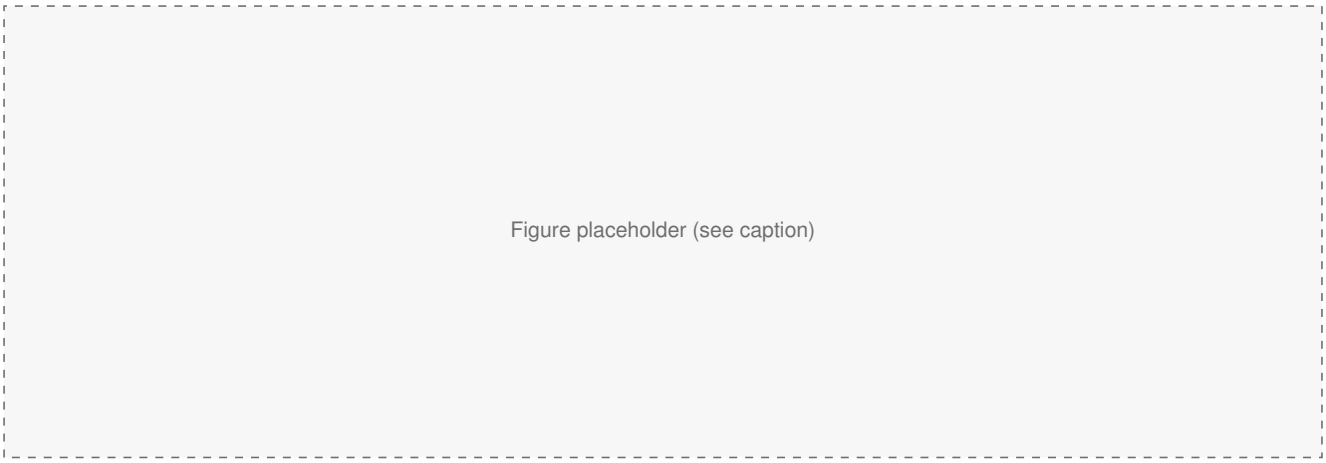


Figure 3. **SAM 3D completes occluded objects from context.** Reconstructions of objects in natural images, rendered from the input view and from a new view. **[TODO: Figure plan: 4 rows (heavy occlusion, truncation by the frame, clutter, small distant object) $\times$ columns: input with prompt, [TODO: 3 baselines], SAM 3D; each method shown from the input view and a 90° rotated view; one inset per row on the hidden part.]**

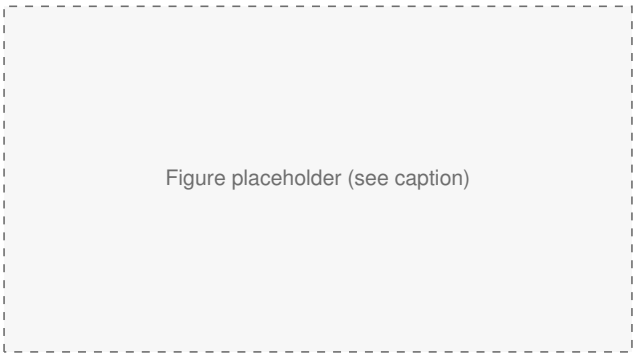


Figure 4. **More data-engine data, better reconstructions. [TODO: Figure plan: x: number of annotated objects or data-engine round (log scale if it spans orders of magnitude); y: win rate against a fixed baseline or Chamfer distance on our benchmark; one curve for SAM 3D, a horizontal line for synthetic-only training; plotted with plot_style.py.]**